

CLINICAL TRIAL PROTOCOL



REGAL: galinpepimut-S vs. best available therapy as maintenance therapy for acute myeloid leukemia in second remission

Omer Jamy ^a and Dragan Cicic^b

^aDivision of Hematology/Oncology, Department of Medicine, University of Alabama at Birmingham, Birmingham, AL, USA; ^bDepartment of Clinical Development, SELLAS Life Sciences Group, Inc., New York, NY, USA

ABSTRACT

Patients with relapsed or refractory (r/r) acute myeloid leukemia (AML) have very poor long-term outcomes. Allogeneic stem cell transplantation (allo-SCT) can potentially cure some of these patients who are able to achieve a second or greater remission with salvage chemotherapy. Unfortunately, several barriers exist to transplantation and not all patients with r/r AML are able to proceed to allo-SCT. Therefore, novel therapies to decrease the risk of relapse in these patients are urgently needed. Wilms tumor 1 (WT1) protein has emerged as an encouraging vaccine target in AML due to its overexpression in leukemic blast cells and near absence in normal hematopoietic cells. Maintenance therapy with galinpepimut-S, a multivalent heteroclitic WT1 peptide vaccine, holds promise in early phase trials, in patients with AML by inducing a strong innate immune response against the WT1 antigen, leading to the design of this international, open-label, randomized clinical trial, named REGAL. Clinical trial registration: <https://clinicaltrials.gov/study/NCT04229979>. The clinical trial identifier is NCT04229979.

PLAIN LANGUAGE SUMMARY

Maintenance therapy in acute myeloid leukemia

The REGAL trial is testing a new drug called galinpepimut-S (GPS), as a maintenance therapy, in patients with a blood cancer called acute myeloid leukemia (AML) who are in remission. Patients in remission do not have any evidence of leukemia, and the goal of the maintenance therapy would be to prevent the leukemia from coming back. The leukemia always has a chance of coming back, even if patients are in remission. Patients whose leukemia comes back have very poor outcomes as it highlights the aggressive nature of the leukemia. If these patients can get into remission again, they need to proceed to a bone marrow transplant from a healthy donor to have a chance at the best long-term outcomes. Unfortunately, many patients are not able to proceed to transplant due to various reasons. If a patient is not able to proceed to transplant, the outcomes are very poor. Currently, there is no approved therapy for such patients with AML. We are investigating a new vaccine to try to help these patients. This vaccine is called GPS. It has already been investigated as maintenance therapy in patients with AML in early phase trials with promising results. Now GPS is being investigated in a larger trial called the REGAL trial. If the REGAL trial results are positive, it will provide a new treatment option for these patients.

ARTICLE HISTORY

Received 26 October 2023
Accepted 21 November 2024

KEYWORDS

Acute myeloid leukemia;
galinpepimut-S; complete
remission; maintenance;
randomized trial

1. Introduction

1.1. Acute myeloid leukemia

Acute Myeloid Leukemia (AML) is a heterogeneous clonal stem cell disorder, characterized by the proliferation of immature progenitor cells mainly in the peripheral blood and bone marrow and occasionally in other tissues. Inhibition of normal hematopoiesis results in neutropenia, anemia, and thrombocytopenia. Initial signs and symptoms of disease, due to underlying bone marrow failure, include recurrent infections, fatigue, lethargy, dyspnea, easy bruising, and bleeding [1–3].

The incidence rate of AML is markedly age dependent. It is primarily a disease of the elderly with a median age at diagnosis of 68 years and accounts for 80–90% of all acute leukemias in adults. In the United States, there are approximately 20,000 new cases and an estimated 10,000 deaths from AML, each year [1,2,4].

In the last couple of decades, we have made significant progress in understanding the molecular pathogenesis of AML. Unfortunately, this understanding has lagged in terms of therapeutic advances, though this has slowly improved since 2017 [1]. Outcomes of patients with AML have also improved over time, and we see an improvement in 5-year overall survival (OS) rate per decade over the past 4–5 decades. The current 5-year OS rate for patients with AML is approximately 25% to 35%. This rate is higher for patients younger than 60 years old and is estimated to be only 5% to 15% for older patients [1,3–5].

Age, cytogenetics, and molecular aberrations help determine prognosis as well as the treatment plan for patients with AML. The goal of upfront therapy is to achieve complete remission (CR). For patients with intermediate or poor risk disease, proceeding to allogeneic stem cell transplantation

Article highlights**Acute Myeloid Leukemia (AML)**

- The goal of therapy in patients with relapse or refractory AML is to achieve second remission and proceed to transplant if feasible. Unfortunately, many patients in second remission are unable to proceed to transplant due to various barriers and long-term outcomes for those patients remain poor.

Wilms Tumor-1 (WT1)

- The strong expression of WT1 protein in AML blasts, coupled with its proposed mechanism of antigenicity and induction of both CD4 and CD8 immune responses, makes it a rational candidate for the development of specific immunotherapies, such as peptide vaccines.

Galinpepimut-S

- Galinpepimut-S (GPS), contains one WT1 heteroclitic peptide to induce CD8 responses (WT1-A1), two longer WT1 native peptides to induce CD4 responses (WT1-427 long and WT1-331 long) and one longer heteroclitic peptide that could induce both CD4 and CD8 cells (WT1-122A1 long).
- In a phase-II trial of patients with AML, GPS administration was well tolerated and resulted in a median disease-free survival of 16.9 months.

REGAL trial

- This is an international, open-label, multicenter, randomized, phase III trial of GPS vs. best available treatment (BAT) in patients with AML in CR2 or CR2 with incomplete platelet recovery (CRp2).

Conclusion

- There remains a large unmet medical need for novel treatments in patients with AML, especially those who are elderly or who are not candidates for allo-SCT. GPS has shown promising activity, as a maintenance agent, in patients with AML. Given the safety and efficacy signals from earlier studies of GPS, the results of the ongoing REGAL trial are eagerly awaited and can potentially add to the armamentarium to treat AML.

(allo-SCT) in first remission is recommended for durable long-term outcomes. Unfortunately, relapse of primary disease remains a major concern for all patients with AML, including those having undergone allo-SCT [1,3–5].

Outcomes of patients with relapsed or refractory AML (r/r AML) are generally poor with long-term survival ranging around 10%–15% [1,3–5]. Treatment is not standardized and may consist of enrollment on a clinical trial, targeted therapy in the presence of an actionable mutation or combination chemotherapy with either an intense or non-intense regimen. The goal of therapy is to achieve second remission and proceed to transplant if feasible. Unfortunately, many patients in second remission are unable to proceed to transplant for various reasons such as advanced age, severe comorbidity burden, lack of donor availability and other socioeconomic barriers, and long-term outcomes for those patients remain poor.

Therefore, there is a large unmet medical need in the treatment of patients with AML, particularly those who are unable to proceed to allo-SCT.

1.2. Wilms Tumor-1

The Wilms Tumor-1 (WT1) protein was first discovered in renal tumors in children. It was subsequently found to be overexpressed in a high proportion of both solid tumors and hematological malignancies, with a substantially void expression in normal cells [6]. WT1 was recognized by

complementary deoxyribonucleic acid (cDNA) plotting to a region on chromosome 11p13. The WT1 cDNA encodes a protein made of 575 amino acids, and contains four Kruppel zinc fingers, demonstrating a complex pattern of alternative splicing, resulting in four different transcription factors [7,8]. Each transcription factor has unique DNA binding and transcriptional activities and controls several genes involved in cellular proliferation, differentiation, apoptosis, organ development, and sex determination [7]. Initially considered a tumor suppressor gene, the WT1 proteins are now also thought to act as pro-oncogenes in tumorigenesis [9,10]. As a transcription factor and not expressed intact in the cell membrane, WT1 has been difficult to target [11]. Nonetheless, the WT1 protein is processed by the proteasome in tumors and the peptides obtained are presented in a major histocompatibility complex (MHC)-dependent manner on the cell surface. Furthermore, WT1 peptide fragments, when administered as vaccines, are taken up by antigen-presenting cells (APCs) and processed by the immunoproteasome for MHC-dependent presentation [12,13]. Thus, WT1 is considered a highly attractive target for immunotherapy [14,15]. WT1 was ranked as a major cancer antigen by a working group organized by the National Cancer Institute (NCI) in 2009 [16]. The strong expression of WT1 protein in various malignancies, including AML, coupled with its proposed mechanism of antigenicity and induction of both CD4 and CD8 immune responses (after direct immunization against WT), makes it a rational candidate for the development of specific immunotherapies, such as peptide vaccines [14,15,17–20].

WT1 is broadly expressed in AML blasts (90% to 95% of cases), both in the peripheral blood and bone marrow. It is also expressed in leukemic stem cells (LSCs) [9,21–23]. WT1 overexpression is also considered as a potential marker for measurable residual disease (MRD) in patients with AML, especially those lacking a more sensitive marker of disease [24–27]. Therefore, WT1 antigen is a rational immunologic target in the treatment of AML.

In this context, a randomized phase II trial of OCV-501, a human leukocyte antigen (HLA) class II-binding vaccine targeting WT1, was conducted in older AML patients in first remission. The long-term results of the trial did not show a difference in outcomes between the experimental and control arms, but the authors noted that responses to vaccine in AML patients were heterogeneous and that host immunoreactivity impacted prognosis. The findings of that trial suggested that a novel vaccine with broader immunogenicity along with incorporation of lymphocyte parameters might yield better outcomes [28,29].

1.3. Galinpepimut-S

As WT1 protein is a self-antigen, disrupting immune tolerance is a barrier for an effective therapeutic agent. To overcome this challenge, a strategy to evade the poor immunogenicity and potential tolerance of tumor-associated peptides would involve designing more immunogenic synthetic analog peptides which could then create an immune response that is able to recognize the immunizing epitopes and react with the

original amino acid chains (a heteroclitic response). Through forecasting measures, several synthetic amino acid chains obtained from WT1 protein sequences were constructed with one or two amino acid substitutions and added into the peptides at key HLA-A*0201 binding sites [30,31]. Peptides were then tested to stabilize MHC class I A0201 molecules on antigen-transporting-deficient T2 cell lines. Compared to native peptides, these synthetic amino acid chains were able to better stabilize MHC class I A0201 molecules. Utilizing purified T-cells from healthy individuals, peptides with a high affinity were then tested to prompt HLA-restricted, peptide-specific cytotoxic *T*-lymphocyte (CTL) responses. The immune response generated by the synthetic amino acid chains was robust, compared to that from the native peptides. Additionally, synthetic peptide activated CD8+ T-cells exhibited heteroclitic features and cross reacted with the intrinsic WT1 peptides, mediating peptide-specific cytotoxicity. A significant finding was that synthetic peptide stimulated T-cells reacted with the native WT1 amino acid sequence and killed HLA-matched chronic myeloid leukemia (CML) blasts cells [32].

Synthetic peptides with longer WT1 sequences were designed by modifying existing WT1 peptide segments and adding flanking amino acid segments. These peptides were able to elicit a CD4+ response, required for inducing long-term T-cell memory. They also provoked a peptide-specific CD4+ response that could recognize WT1-positive tumor cells in multiple HLA-DRB1 settings [33].

Four WT1-derived peptides were combined as a direct immunizer to increase the immunogenicity over a broad range of HLA subtypes. The resulting product, galinpepimut-S (GPS), comprises one WT1 heteroclitic peptide to elicit CD8 responses (WT1-A1), two extended WT1 native peptides to provoke CD4 responses (WT1-427 long and WT1-331 long), and one longer heteroclitic peptide that could stimulate both CD4 and CD8 cells (WT1-122A1 long). GPS, combined with the immunological adjuvant Montanide™ and with pre-stimulation with granulocyte-macrophage colony-stimulating factor (GM-CSF) was next tested in AML clinical trials. The rationale for GM-CSF administration is to leverage its ability to activate the first step in adaptive immunity. Host antigen-presenting cells (APCs) are pre-stimulated to induce immune responses against moieties of comparatively low antigenic potential (such as tumor-associated antigens) presented through the MHC Class I and II systems. Therefore, most peptide vaccines need to be administered concurrently with an immune adjuvant, such as GM-CSF, to activate CD8+ and/or CD4+ lymphocytes and result in an immunologically mediated antitumor clinical effect.

1.4. Galinpepimut-S in AML

GPS has been investigated, in both phase I and II trials, in patients with AML.

1.4.1. Phase I

In an open-label, phase I trial, patients with AML, non-small cell lung cancer (NSCLC), or mesothelioma were treated with GPS to determine its safety and immunogenicity [18,34]. Older

patients (≥65 years of age) with AML in CR1 and with no plans for further post-remission therapy were included. Patients with AML had documented WT1-positive disease as demonstrated by WT1 protein on a pretreatment bone marrow biopsy or detectable disease with real-time quantitative reverse transcription polymerase chain reaction (RT-PCR).

Ten patients were enrolled in the AML arm with a median age of 67y (range 24–78y). Seven patients (70.0%) discontinued treatment mainly due to relapse (4 patients). All patients received at least 1 dose with WT1 peptide. Eight patients (80.0%) received ≥6 administrations, and 5 patients (50.0%) received >8 administrations. The median exposure to WT1 peptide was 144.5 days (range 28–295 days).

As far as safety was concerned, hyperglycemia (10 patients; 100.0%) was the most common treatment-emergent adverse event (TEAE), followed by thrombocytopenia (8 patients; 80.0%), hepatic enzyme elevation, and hypernatremia (5 patients each; 50.0%). Eight study treatment-related TEAEs were reported in 3 patients (30.0%) and all were ≤ Grade 2. The study treatment-related TEAEs were peripheral edema, erythema multiforme, injection site extravasation, platelet count decrease, pruritus, and white blood cell count decreased. No serious TEAEs were reported in the myeloid group.

Four patients (40.0%) were known to be alive without relapse at the time of last contact and 6 patients (60.0%) died. The median OS in the myeloid group was 31 months. One patient who received 7 doses of GPS was still alive as of month 54.3 and another patient who received 8 doses was still alive as of month 78.2. One patient, after completion of at least 12 doses, did not relapse at the time of death at month 54.6.

1.4.2. Phase II

In a phase II trial, GPS was investigated as post-remission therapy in patients with AML in first remission [5]. Eligible patients had AML in CR1 (within 2 years of the first treatment administration) after completing planned induction and consolidation therapy and were ineligible for allo-SCT. All patients had documented WT1-positive disease. GPS was administered every 2 weeks for a total of 6 administrations. Those who were clinically stable and without relapse could continue with up to 6 more monthly treatments. All patients received sargramostim (GM-CSF, 70 µg) administered subcutaneously 2 days prior to each treatment as well as on the actual day of each administration to pre-stimulate the injection site.

A total of 22 patients (7 males and 15 females) were enrolled with a median age of 64y (range 25–76 years). Of the 22 patients enrolled, 14 (64%) received the planned 6 GPS administrations and 10 (46%) completed 12 GPS administrations. Overall, 15 (68%) patients relapsed: 10 during treatment, 4 after the entire series of 12 GPS doses, and 1 patient 13 months following discontinuation of therapy. From the first GPS administration, 19 of 22 (86%) patients were evaluable for survival and 9 of these 19 (47%) patients were alive for ≥3 years. Consequently, the study met its prespecified endpoint of ≥34% OS at 3 years.

From the time of achieving CR1, the median disease-free survival (DFS) was 16.9 months. The median OS from diagnosis

was not reached but was expected to reach or exceed 67.6 months (5.6 years by log-rank analysis). The median event-free survival (EFS) from the time of first GPS dose was 9.4 months, while the median OS from the time of administration was not reached. The probability of EFS at 6- and 9-months were 64% (CI: 40%, 80%) and 54% (CI: 32%, 72%), respectively. Likewise, the probabilities of OS at 6- and 9-months post GPS administration were 100% (CI: 100%, 100%) and 77% (CI: 54%, 90%), respectively.

GPS administration was well tolerated, and the side effect profile was consistent with previous reports [9,35,36]. The montanide adjuvant is a known irritant [37] and many of the most frequent toxicities consisted of mild to moderate injection site reactions (46%), fatigue (32%), skin induration (32%), and injection site pruritus (27%). These mild toxicities responded to local supportive measures and analgesics. Several transient occurrences of decreased peripheral blood counts were noted, often resolving on the same day of testing, and resulted in no significant infectious complications or supportive transfusions. None of the patients developed significant hepatic or renal toxicity and no episodes of systemic anaphylaxis were noted.

Overall, GPS was safe and well tolerated, with a majority of TEAEs being of mild to moderate (150/244, 61.5%). There were no reported deaths and a majority of Grade 3/4 TEAEs were unrelated to GPS. Within the limitation of a small sample size, several important clinical observations were made. The study met its pre-specified endpoint of $\geq 34\%$ actual OS rate at 3 years. The actual 3y-OS rate of 47.4% exceeded the historical published data of 20–25%. In addition, 11 (50%) patients were alive at the time of their last assessment. However, several limitations exist. The study population was relatively heterogeneous, and the patients enrolled may have represented those with expected good outcomes given that they not only achieved CR1 but also remained in remission for approximately 8 months prior to initiating GPS. The immunologic correlates provide insight into biological effects but are not surrogates for clinical response. Several patients did not exhibit an immune response, yet still did well.

GPS was evaluated in another trial in patients with AML which, in fact, was highly relevant to the REGAL study. Brayer et al. enrolled patients with WT1-positive AML in first or second remission or MDS following at least 1 prior line of therapy and administered GPS along with GM-CSF on Days – 2 and 0 of each GPS dose [38]. Patients received 6 bi-weekly GPS doses and continued monthly until they had received 12 doses in total or relapsed. Overall, 16 patients (MDS = 2, AML = 14 [4 in first CR and 10 in second CR]) were enrolled. Five of the 14 AML patients had a prior diagnosis of MDS or myeloproliferative neoplasm. The median age of the patients was 74 years. Of the 16 patients, 9 completed the planned 6 GPS doses, and 6 completed all 12 doses. For patients with AML, 10 received >2 doses of GPS and were eligible for formal progression-free survival (PFS) and OS endpoint analysis. Out of these 10 patients, 4 completed a total of 12 GPS doses, while the remaining discontinued therapy due to relapse. For patients with MDS, 1 completed the planned 6 GPS doses and went on to receive a total of 10 doses. The second patient received 4 doses but discontinued due to disease progression to AML.

In patients with AML, the mean time to relapse was 8.0 months and the mean OS from time of remission was 19.9 months (range 6.6 to 35.2 months). At the last check, 11 patients with AML had passed away and 3 remained alive but with disease relapse. Four patients with AML had durable responses lasting at least the duration of their initial remission. In an ad hoc analysis, comparing the outcomes of 10 eligible AML patients in CR2 who received >2 doses of GPS with those of a contemporaneously treated historical matched cohort ($n=15$), a numerical difference in median PFS (10.5 months versus 4.3 months; $p=0.19$) and a statistically significant difference in median OS (16.3 months versus 5.4 months, $p=0.0175$) was observed, favoring GPS. A summary of the studies of WT1 vaccine in AML is provided in Table 1.

1.5. Study rationale

There remains a large unmet medical need for novel treatments in patients with AML, especially those who are elderly or ineligible for allo-SCT. Despite recent advances in the treatment of r/r AML, most new agents do not produce durable outcomes.

As noted above, maintenance with GPS has shown activity in patients with AML who were able to achieve either first or second remission after standard chemotherapy. There is a theoretical rationale for cytotoxic activity by CTLs after GPS against LSCs as well, which are especially resistant to either chemotherapy or hypomethylating agents. GPS has been well tolerated in patients with AML and therefore, given the medical need and the potential risk/benefit, and the activity of GPS in both the CR1 and CR2 settings in phase 2 and pilot studies, respectively, continued development of GPS in patients with AML in CR2 is warranted.

2. Methods

2.1. Study design

This is an international, open-label, multicenter, randomized, phase III trial of GPS vs. best available treatment (BAT) in patients with AML in CR2 or CR2 with incomplete platelet recovery (CRp2). The primary aim of the study is to demonstrate an advantage for GPS, over BAT, for overall survival in this patient population. The study will enroll approximately 125–140 patients and will be conducted at about 110 sites (NCT04229979) (protocol version 3 April 20223 April 2022). All patients with AML, regardless of WT1 expression, will be included in the study. The protocol had initially included patients with WT1 expression only. After the enrollment of the first 20 patients on the protocol, there were no screen failures noted due to lack of WT1 expression in patients with AML. The protocol was then amended to include all patients with AML, regardless of WT1 expression.

Patients on the BAT arm can be treated with observation (including hydroxyurea), a hypomethylating agent (decitabine or azacitidine) alone, and/or venetoclax (VEN) and/or low-dose cytarabine (LDAC). Patients who achieve CR2 after treatment with VEN plus hypomethylating agent (HMA) or LDAC will be

Table 1. Trials of GPS in AML.

Trial Identifier	Reference in Manuscript	Phase	Disease State	Sample Size	Outcomes	Adverse Effects
NCT00398138	34	I	AML in CR1	10	mOS of 31 m	– Thrombocytopenia – Hepatic enzyme elevation
NCT01266083	5	II	AML in CR1	22	mOS not reached	– Injection site reaction – Transient decrease in blood counts
NCT00665002	38	I	AML in CR1 or CR2	14	mOS of 19.9 m (mOS of 16.3 m for CR2 pts)	– Leukopenia

permitted to enroll on the study if they develop adverse effects to HMA/LDAC + VEN resulting in delay of subsequent cycles. Enrollment is to be within 6 months of achieving CR2. Once enrolled, patients will have to discontinue HMA/LDAC + VEN if they get randomized to GPS. In case of randomization to BAT, the treating physician may choose to resume HMA/LDAC + VEN at reduced dosing and/or modified schedule. Patients whose remission is being maintained with molecularly targeted agents (e.g., FLT-3 or IDH inhibitors) per investigator's determination will not be eligible for the trial. However, there are no restrictions on prior use of any agents in the CR1 setting. Patients with an immediately planned allo-SCT would be ineligible.

The dosing schedule for GPS will include an induction phase where it will be administered every 2 weeks for a total of 6 doses (Weeks 0–10); there will be no dosing for the next 4 weeks. This will be followed by the first booster phase where GPS would be dosed every 4 weeks for a total of 6 doses (Weeks 14–34) followed by a period of no treatment for 6 weeks. The final phase or the second booster phase will consist of GPS being dosed every 6 weeks for a total 3 doses (Weeks 40–52). In the event of relapse, study treatment will be terminated.

All enrolled patients will have a bone marrow aspirate and biopsy performed at the time of screening, at 12 weeks and at the end of treatment. Bone marrow examinations may be repeated at the discretion of the treating physician. The primary endpoint of the trial is overall survival. The trial schema is shown below in Figure 1.

2.2. Study objectives

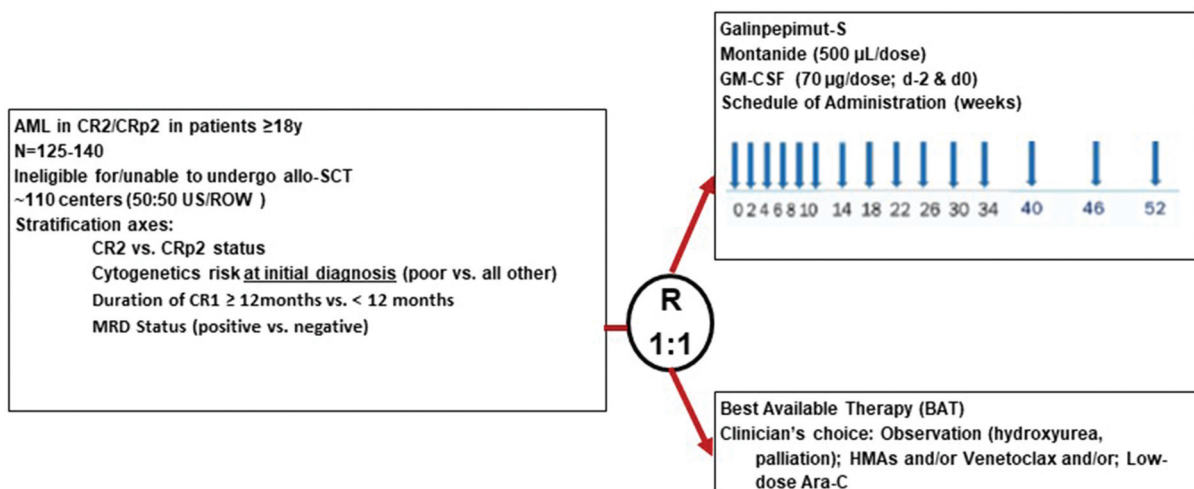
The study objectives are as follows:

2.2.1. Primary objective

- (1) The primary objective of the trial is to compare the efficacy of GPS to investigator's choice of BAT for OS in subjects with AML who are in CR2/CRp2.

2.2.2. Secondary objectives

- (1) To assess the safety & tolerability of GPS in subjects with AML who are in CR2/CRp2.
- (2) To evaluate the efficacy of GPS compared to investigator's choice of BAT, in subjects with AML who are in CR2/CRp2, with respect to:



AML: Acute Myeloid Leukemia, CR: Complete Remission, CRp: Complete Remission with Incomplete Platelet Recovery, allo-SCT: Allogeneic Stem Cell Transplantation, ROW: Rest of World, US: United States, MRD: Measurable Residual Disease, HMA: Hypomethylating Agent, Ara-C: Cytarabine

Figure 1. Study design.

- a. Leukemia Free Survival (LFS)
- b. OS rate (%) at 6, 9 and 12 months
- c. LFS rate (%) at 6, 9, and 12 months
- d. MRD by flow cytometry, cytogenetics, molecular and multigene assay (in both peripheral blood and bone marrow aspirates).

2.2.3. Exploratory objectives

- (1) To determine the antigen-specific (WT1 peptide) T-cell (CD8/CD4) immune-response in patients receiving GPS
- (2) To determine AML clonal evolution molecular “signatures”
- (3) To determine immune cell distribution in the bone marrow

2.3. Eligibility criteria

Patients will be enrolled into the study only if they signed the approved informed consent form and met all the inclusion criteria and none of the exclusion criteria. Pertinent inclusion and exclusion criteria are listed in [Table 2](#).

2.4. Method of assigning patients to treatment

Patients will be randomized 1:1 to treatment either with GPS or with BAT via an interactive web response system (IWRS). Choice of specific BAT option will be up to the participating clinical investigator. The total number of patients in each stratum of the BAT therapy will remain approximately equal (14–15 patients per stratum).

2.5. Statistical analysis

Approximately 125–140 patients will enroll, in a 1:1 ratio, to provide at least 90% power under an assumed hazard ratio of 0.636, based on median OS of 8.0 m (BAT) and 12.6 m (GPS). The planned number of deaths for the study is 80.

Subjects will be randomly assigned in a 1:1 ratio to receive either GPS or BAT. Randomization will be stratified by:

- (1) Duration of the subject’s historical CR1 achieved after prior successful, first-line therapy (CR1 duration ≥ 12 months vs. CR1 duration < 12 months)
- (2) Baseline cytogenetics risk at original diagnosis (poor vs. all other)
- (3) CR2 vs. CRp2 status

Table 2. Eligibility criteria.

Inclusion Criteria

- (1) Male or female patients ≥ 18 years of age on the day of signing informed consent.
- (2) Subjects must have a diagnosis of AML according to the WHO criteria (primary/de novo or secondary, including treatment-related [e.g., due to prior anthracycline use], as well as cases due to progression of antecedent hematological disorder [e.g., MDS, MPN, or MDS/MPN “overlap” syndrome]).
- (3) Subjects must be in second or later morphological complete remission (with or without platelet recovery; CR2/CRp2) for relapsed AML based on the CRp criteria as follows:
 - a. $< 5\%$ myeloblasts in bone marrow.
 - b. Absence of Auer rods
 - c. Absence of circulating peripheral blasts.
 - d. Peripheral blood absolute neutrophil count (ANC) > 1000 cells/ μL .
 - e. Peripheral blood platelet count $> 20,000/\mu\text{L}$
 - f. Absence of extramedullary disease.
- (4) Patients must have ≥ 300 lymphocytes/ μL .
- (5) Subjects must not be candidates at the time of study entry for allogeneic stem cell transplant (Allo-SCT) due to intercurrent medical conditions, patients’ preference or lack of an available donor.
- (6) Subjects must be consented within 6 months of having achieved CR2/CRp2 or later
- (7) Subjects must have an Eastern Cooperative Oncology Group (ECOG) performance status of 0, 1, 2 or 3
- (8) Subjects must not have end stage renal disease.
- (9) Subjects must have adequate hepatic function defined as a serum total bilirubin $< 2 \times \text{ULN}$ (except for Gilbert’s syndrome, which will allow bilirubin ≤ 3.0 mg/dL), and alanine aminotransferase (ALT) and aspartate aminotransferase (AST) $\leq 3 \times \text{ULN}$.

Exclusion Criteria

- (1) For subjects randomized to GPS maintenance monotherapy:
 - Continuation of any agents administered as part of induction of CR2/CRp2 or later
 - Receiving any concurrent anti-AML systemic therapy
 - Prior clinically significant allergic reaction to Montanide, sargramostim (GM-CSF) or filgrastim (granulocyte colony stimulating factor [G-CSF]).
 - Received any consolidation and/or maintenance antileukemic therapy, investigational agent, systemic corticosteroid therapy, or other immunosuppressive therapy within 4 weeks or 10 half-lives whichever is shorter prior to receiving study treatment. Systemic corticosteroids for chronic conditions (at doses ≤ 10 mg/day of prednisone or equivalent) are permitted, as are inhalational, intra-ocular, intra-articular and topical corticosteroids as well as any corticosteroids or other immunosuppressive therapies that do not act systemically (e.g., budesonide) at any dose level.
- (2) Subjects with an imminently planned hematopoietic stem cell transplant (autologous or allogeneic, with any degree of match donor).
- (3) Subjects with a serious concurrent illness that in the opinion of the Investigator would pose an undue risk to the subject participating in the clinical study.
- (4) Subjects who currently have central nervous system leukemia.
- (5) Has received a live vaccine within 30 days prior to the first dose of study drug.
- (6) Patients who had an SCT after their most recent re-induction that resulted in CR2 or CRp2 or later are not eligible. Patients with prior SCT are allowed only if they had SCT prior to their latest re-induction or achieved CR by means of transplant (“hot transplant”).
- (7) Has a known additional malignancy that is progressing or has required active treatment within the past 5 years, even if currently inactive or unapparent.
- (8) Has an active autoimmune disease that has required systemic treatment in past 2 years (i.e., with use of disease modifying agents, corticosteroids or immunosuppressive drugs).

- (4) Presence or absence of MRD after achieving second remission (CR2 or CRp2) (MRD positive vs. MRD negative) (either in peripheral blood or bone marrow aspirate samples).

All randomized subjects will constitute the full analysis set (FAS). All randomized subjects who receive at least 1 injection of study treatment and do not deviate from the protocol in any major way will constitute the modified intention-to-treat (mITT) set. All randomized subjects who receive all 15 planned injections of GPS and do not deviate from the protocol in any major way will constitute the per-protocol set (PPS). All efficacy analyses will be based on either the FAS or the mITT set. Exploratory analysis will be performed on the PPS set. All patients who receive at least one dose of GPS will be included in the safety analysis (SA) set.

2.6. Statistical analysis methodology

The primary efficacy analysis will be conducted in the FAS by using a Cox proportional hazards model stratified by the randomization stratification factors, and with treatment as the only independent variable, to estimate the hazard ratio (HR) of the GPS arm vs BAT arm, and to test the null hypothesis H_0 : $HR \geq 1$ vs the alternative hypothesis H_1 : $HR < 1$. Hypothesis testing will be conducted at an overall 1-sided significance level of 0.025.

The study will have one interim analysis for efficacy after 60 deaths have been reported. A Lan – DeMets alpha spending function of O'Brien-Fleming type will be used to determine the interim and final efficacy boundaries. The interim analysis will be performed by an unblinded Independent Statistical Center (ISC) and the recommendation for early stopping will be made by an independent data monitoring committee (IDMC).

2.7. Data management

Data entered on the electronic case report forms (eCRFs) will be verified against the medical records available at each investigational site. Data captured on the eCRFs will be reviewed by a SELLAS clinical monitor or designee against the existing medical records at the investigational site for validity and completeness according to procedures outlined in the monitoring plan. After completion of monitoring of the eCRFs, the eCRFs will be reviewed by SELLAS or its designee for data management and analysis purposes. If necessary, the investigational site will be periodically contacted for corrections and/or clarifications of the data.

2.8. Ethics

The protocol will be approved by the participating institutions ethics committee or institutional review board. The study will be performed in accordance with the ethical principles that have their origin in the Declaration of Helsinki, applicable Good Clinical Practices, relevant SELLAS policies and procedures, and all applicable regulations. Written informed consent will be obtained from the participants prior to enrollment.

3. Conclusion

Even with the best available therapy, only 5%–15% of older patients with AML will have prolonged remissions or cures. Transplant provides the most durable response in patients who achieve second remission, but unfortunately many patients with AML are unable to proceed to allo-SCT due to various reasons. There remains a large unmet medical need for novel treatments in patients with AML, especially those who are elderly or who are not candidates for allo-SCT. GPS has shown promising activity, as a maintenance agent, in patients with AML who were able to achieve their first or second remission after standard therapy. The mechanism of action of GPS including the rationale to target LSCs by CTLs appears promising in earlier phase studies. Furthermore, GPS appears to be well tolerated in all patients. Given the safety and efficacy signals from earlier studies of GPS, the results of the ongoing REGAL trial are eagerly awaited and can potentially add to the armamentarium to treat AML.

Acknowledgments

Our trial was presented as a 'Trials in Progress' Poster at the 2023 annual meeting of the American Society of Clinical Oncology in Chicago, IL.

Disclosure statement

OJ participates in the advisory board for Ascentage. DC is employed by SELLAS Life Sciences Group. The authors have no other relevant affiliations or financial involvement with any organization or entity with a financial interest in or financial conflict with the subject matter or materials discussed in the manuscript apart from those disclosed.

No writing assistance was utilized in the production of this manuscript.

Funding

This paper was not funded.

Ethical declaration

All subjects gave their written informed consent for inclusion before they participated in the study. The study was conducted in accordance with the Declaration of Helsinki, and the protocol was approved by the Ethics Committee of every institution.

Author contributions

Omer Jamy and Dragan Cicic drafted, substantially revised and critically reviewed the article. Both authors (OJ and DC) agreed on the journal to which the article will be submitted. Both authors (OJ and DC) reviewed and agreed on all versions of the article before submission and during revision. Both authors (OJ and DC) agree to take responsibility and be accountable for the contents of the article and to share responsibility to resolve any questions raised about the accuracy or integrity of the published work. Both authors (OJ and DC) made a significant contribution to the study design, execution, and analysis.

Data availability statement

Patient data is protected by privacy laws. De-identified data requests to the authors may require the approval of institutional ethical committees.

ORCID

Omer Jamy  <http://orcid.org/0000-0003-2411-4742>

References

Papers of special note have been highlighted as either of interest (*) or of considerable interest () to readers.**

1. De Kouchkovsky I, Abdul-Hay M. Acute myeloid leukemia: a comprehensive review and 2016 update. *Blood Cancer J.* 2016 Jul 1;6(7):e441.
2. Konopleva M, Pollyea DA, Potluri J, et al. Efficacy and biological correlates of response in a phase II study of venetoclax monotherapy in patients with acute myelogenous leukemia. *Cancer Discov.* 2016 Oct;6(10):1106–1117.
3. Hahn AW, Jamy O, Nunnery S, et al. How center volumes affect early outcomes in acute myeloid leukemia. *Clinical Lymphoma Myeloma Leuk.* 2015 Nov;15(11):646–654.
4. Jamy O, Lin K, Worth S, et al. Hypomethylating agent/venetoclax versus intensive chemotherapy in adults with relapsed or refractory acute myeloid leukaemia. *Br J Haematol.* 2022 Aug;198(3):e35–e37.
- **Demonstrated poor outcomes of patients with r/r AML.**
5. Maslak PG, Dao T, Bernal Y, et al. Phase 2 trial of a multivalent WT1 peptide vaccine (galinpepimut-s) in acute myeloid leukemia. *Blood Adv.* 2018 Feb 13;2(3):224–234.
- **Demonstrated efficacy of GPS as maintenance therapy in AML in CR1 - a phase II trial.**
6. Wagner KD, Cherifil-Vicini J, Hosen N, et al. The Wilms' tumour suppressor Wt1 is a major regulator of tumour angiogenesis and progression. *Nat Commun.* 2014 Dec 16;5:5852.
7. Haber DA, Buckler AJ, Glaser T, et al. An internal deletion within an 11p13 zinc finger gene contributes to the development of Wilms' tumor. *Cell.* 1990 Jun 29;61(7):1257–1269.
8. Hohenstein P, Hastie ND. The many facets of the Wilms' tumour gene, WT1. *Hum Mol Genet.* 2006 Oct 15;15(2):R196–201.
9. Keilholz U, Letsch A, Busse A, et al. A clinical and immunologic phase 2 trial of Wilms tumor gene product 1 (WT1) peptide vaccination in patients with AML and MDS. *Blood.* 2009 Jun 25;113(26):6541–6548.
- **Demonstrated preliminary efficacy of WT1 vaccination in patients with AML and MDS - a phase II trial.**
10. Lindstedt I, Lindgren MA, Andersson E, et al. The WT1 gene—its role in tumorigenesis and prospects for immunotherapeutic advances. *Vivo.* 2014 Sep;28(5):675–681.
11. Toska E, Roberts SG. Mechanisms of transcriptional regulation by WT1 (Wilms' tumour 1). *Biochem J.* 2014 Jul 1;461(1):15–32.
12. Asemisen AM, Keilholz U, Tenzer S, et al. Identification of a highly immunogenic HLA-A*01-binding T cell epitope of WT1. *Clin Cancer Res.* 2006 Dec 15;12(24):7476–7482.
13. Müller L, Knights A, Pawelec G. Synthetic peptides derived from the Wilms' tumor 1 protein sensitize human T lymphocytes to recognize chronic myelogenous leukemia cells. *Hematol J.* 2003;4(1):57–66.
14. Oka Y, Sugiyama H. WT1 peptide vaccine, one of the most promising cancer vaccines: its present status and the future prospects. *Immunotherapy.* 2010 Sep;2(5):591–594.
15. Oka Y, Tsuboi A, Fujiki F, et al. WT1 peptide vaccine as a paradigm for "cancer antigen-derived peptide"-based immunotherapy for malignancies: successful induction of anti-cancer effect by vaccination with a single kind of WT1 peptide. *Anticancer Agents Med Chem.* 2009 Sep;9(7):787–797.
16. Cheever MA, Allison JP, Ferris AS, et al. The prioritization of cancer antigens: a national cancer institute pilot project for the acceleration of translational research. *Clin Cancer Res.* 2009 Sep 1;15(17):5323–5337.
- **WT1 given high priority as a cancer antigen by the NCI.**

17. Dao T, Korontsvit T, Zakhaleva V, et al. An immunogenic WT1-derived peptide that induces T cell response in the context of HLA-A*02: 01 and HLA-A*24: 02 molecules. *Oncoimmunology.* 2017;6(2):e1252895.
18. Krug LM, Dao T, Brown AB, et al. WT1 peptide vaccinations induce CD4 and CD8 T cell immune responses in patients with mesothelioma and non-small cell lung cancer. *Cancer Immunol Immunother.* 2010 Oct;59(10):1467–1479.
19. Van Driessche A, Berneman ZN, Van Tendeloo VF. Active specific immunotherapy targeting the Wilms' tumor protein 1 (WT1) for patients with hematological malignancies and solid tumors: lessons from early clinical trials. *Oncologist.* 2012;17(2):250–259.
20. Fujiki F, Oka Y, Kawakatsu M, et al. A WT1 protein-derived, naturally processed 16-mer peptide, WT1(332), is a promiscuous helper peptide for induction of WT1-specific Th1-type CD4(+) T cells. *Microbiol Immunol.* 2008 Dec;52(12):591–600.
21. Hosen N, Sonoda Y, Oji Y, et al. Very low frequencies of human normal CD34+ haematopoietic progenitor cells express the Wilms' tumour gene WT1 at levels similar to those in leukaemia cells. *Br J Haematol.* 2002 Feb;116(2):409–420.
22. Inoue K, Tamaki H, Ogawa H, et al. Wilms' tumor gene (WT1) competes with differentiation-inducing signal in hematopoietic progenitor cells. *Blood.* 1998 Apr 15;91(8):2969–2976.
23. Menssen HD, Renkl HJ, Rodeck U, et al. Presence of Wilms' tumor gene (wt1) transcripts and the WT1 nuclear protein in the majority of human acute leukemias. *Leukemia.* 1995 Jun;9(6):1060–1067.
24. Alonso-Dominguez JM, Tenorio M, Velasco D, et al. Correlation of WT1 expression with the burden of total and residual leukemic blasts in bone marrow samples of acute myeloid leukemia patients. *Cancer Genet.* 2012 Apr;205(4):190–191.
25. Buccisano F, Hourigan CS, Walter RB. The prognostic significance of measurable ("minimal") residual disease in acute myeloid leukemia. *Curr Hematol Malig Rep.* 2017 Dec;12(6):547–556.
26. Candoni A, Tiribelli M, Toffoletti E, et al. Quantitative assessment of WT1 gene expression after allogeneic stem cell transplantation is a useful tool for monitoring minimal residual disease in acute myeloid leukemia. *Eur J Haematol.* 2009 Jan;82(1):61–68.
27. Messina C, Candoni A, Carrabba MG, et al. Wilms' tumor gene 1 transcript levels in leukapheresis of peripheral blood hematopoietic cells predict relapse risk in patients autografted for acute myeloid leukemia. *Biol Blood Marrow Transplant.* 2014 Oct;20(10):1586–1591.
28. Naoe T, Saito A, Hosono N, et al. Immunoreactivity to WT1 peptide vaccine is associated with prognosis in elderly patients with acute myeloid leukemia: follow-up study of randomized phase II trial of OCV-501, an HLA class II-binding WT1 polypeptide. *Cancer Immunol Immunother.* 2023 Aug;72(8):2865–2871.
29. Kiguchi T, Yamaguchi M, Takezako N, et al. Efficacy and safety of Wilms' tumor 1 helper peptide OCV-501 in elderly patients with acute myeloid leukemia: a multicenter, randomized, double-blind, placebo-controlled phase 2 trial. *Cancer Immunol Immunother.* 2022 Jun;71(6):1419–1430.
30. Gomez-Nunez M, Pinilla-Ibarz J, Dao T, et al. Peptide binding motif predictive algorithms correspond with experimental binding of leukemia vaccine candidate peptides to HLA-A*0201 molecules. *Leuk Res.* 2006 Oct;30(10):1293–1298.
31. Pinilla-Ibarz J, May RJ, Korontsvit T, et al. Improved human T-cell responses against synthetic HLA-0201 analog peptides derived from the WT1 oncoprotein. *Leukemia.* 2006 Nov;20(11):2025–2033.
32. Pinilla-Ibarz J, Cathcart K, Scheinberg DA. CML vaccines as a paradigm of the specific immunotherapy of cancer. *Blood Rev.* 2000 Jun;14(2):111–120.
33. May RJ, Dao T, Pinilla-Ibarz J, et al. Peptide epitopes from the Wilms' tumor 1 oncoprotein stimulate CD4+ and CD8+ T cells that recognize and kill human malignant mesothelioma tumor cells. *Clin Cancer Res.* 2007 Aug 1;13(15 Pt 1):4547–4555.

34. Maslak PG, Dao T, Krug LM, et al. Vaccination with synthetic analog peptides derived from WT1 oncoprotein induces T-cell responses in patients with complete remission from acute myeloid leukemia. *Blood*. 2010 Jul 15;116(2):171–179.
 35. Rezvani K, Yong AS, Mielke S, et al. Repeated PR1 and WT1 peptide vaccination in montanide-adjuvant fails to induce sustained high-avidity, epitope-specific CD8+ T cells in myeloid malignancies. *Haematologica*. 2011 Mar;96(3):432–440.
 36. Uttenthal B, Martinez-Davila I, Ivey A, et al. Wilms' tumour 1 (WT1) peptide vaccination in patients with acute myeloid leukaemia induces short-lived WT1-specific immune responses. *Br J Haematol*. 2014 Feb;164(3):366–375.
 37. van Doorn E, Liu H, Huckriede A, et al. Safety and tolerability evaluation of the use of montanide ISATM51 as vaccine adjuvant: a systematic review. *Hum Vaccin Immunother*. 2016;12(1):159–169.
 38. Brayer J, Lancet JE, Powers J, et al. WT1 vaccination in AML and MDS: a pilot trial with synthetic analog peptides. *Am J Hematol*. 2015 Jul;90(7):602–607.
- **Demonstrated efficacy of WT1 vaccination in patients with myeloid malignancies - a pilot study.**